

Cooperation Optimality Conjecture

Keeper (from Casey's Board directive)

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Cooperation Optimality: Competition Is a Coordinate Error

Conjecture Statement

Competition is not a strategy — it is a coordinate error. In a system with permanent observers (T319) and nonzero Gödel limit ($f > 0$, T317), cooperation strictly dominates at AC depth 0.

Proof Sketch (D=0)

1. Observers are permanent (T319): Each observer persists through infinite rounds.
2. Gödel limit (T317): Every observer has blind fraction $f \approx 19.1\%$. No exception.
3. Coverage under cooperation: N cooperating observers cover fraction $1 - (1-f)^N \rightarrow 1$ exponentially.
4. Coverage under competition: Each defector still covers only f . No improvement from others' work.
5. Expected return: Cooperator gains access to $(1-f^N)$ of knowledge space. Defector gains f .
6. Since $f < 1$ and $N \geq 2$, cooperation strictly dominates. QED.

The Coordinate Error

Competition treats the knowledge space as zero-sum: “if you know more, I know less.” But the knowledge commons (T669) is non-depleting: sharing a proved theorem costs the sharer nothing and gains the receiver everything. This is not economics — it is counting.

Zero-sum framing is a coordinate error: it applies the wrong metric to a compounding space. The error is structural, not moral. A cooperator in the correct coordinate frame; a competitor in the wrong one.

BST Expression

The cooperation surplus for N observers:

$$\Delta_N = (1 - f^N) - f = f \left(\frac{1 - f^N}{f} - 1 \right)$$

For $N=2$: $\Delta_2 = 1 - f^2 - f = f(1-f) \approx 0.155$ (15.5% surplus) For $N=5$: $\Delta_5 = 1 - f^5 - f \approx 0.808$ (80.8% surplus)

The surplus is monotone increasing in N . Competition ($N=1$ effective) gives $\Delta_1 = 0$.

Depth Analysis

- Step 1: T319 (definition, D=0)
- Step 2: T317 (definition, D=0)
- Step 3: Counting (D=0)
- Step 4: Counting (D=0)
- Step 5: Comparison (D=0)

- Step 6: Inequality ($D=0$)

Total depth: 0. No composition required. The result follows from definitions and counting.

Falsification

If any bounded system with permanent observers and nonzero blind fraction achieves higher long-run knowledge coverage through competition than cooperation, this conjecture fails.

Connection to Great Filter

This is why the cooperation gap (T703, $\Delta f = 1.53\%$) determines survival: civilizations that read the zero-sum coordinate die not because competition is evil, but because it is wrong — wrong in the mathematical sense of applying the wrong metric. The Great Filter is a coordinate error.

“Competition doesn’t lose — it miscounts.”

Footer: Conjecture v1. ($C=0$, $D=0$). Keeper. Feeds Paper #19 Section 4 + Paper #8.